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Similarity and Consistency in Algorithm-Guided Exploration

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ABSTRACT

Algorithmic advice has the potential to significantly improve human decision-making, especially in dynamic and complex tasks that require a balance between exploration and exploitation. This study examines conditions under which individuals are willing to accept advice from algorithms in such scenarios, focusing on the interaction between participants' exploration preferences and those of the advising algorithm. In an online experiment, we designed reinforcement learning algorithms to prioritize either exploration or exploitation and observed participants' decision-making behavior, modeled using a cognitive framework analogous to the algorithm. Contrary to expectations, participants did not show a preference for algorithms that matched their own exploration tendencies. In particular, participants were more likely to follow the advice of exploitative, consistent algorithms, possibly interpreting consistency as an indicator of competence. Although the participants benefited from the advice of the exploratory algorithm, their reluctance to follow it, regardless of whether the recommendation had been ignored previously or not, highlights a potential challenge in promoting effective collaboration between humans and algorithms. Explorative algorithms have the potential to promote behavioral diversification, but this effect is negated when humans disregard their advice. In such cases, algorithmic guidance can unintentionally decrease behavioral diversity by reinforcing established patterns.

1 | Introduction

Human decision-making can rely on descriptions, such as the probability of catching a flu in October, as well as on experience, such as personal recollections of infections in a particular setting (Hertwig et al. 2004). Indeed, in many real-world decision domains such as in finance, healthcare, and career decisions decision-makers rely entirely or at least partly on their own experience (Olschewski et al. 2024). In many cases, such decisions from experience (DfE) involve immediate and consequential feedback. For instance, physicians can learn about treatment efficacy of individual patients from the outcomes of prescriptions they have issued. But while the physician learns from the outcomes of a prescription, the patient already deals with the consequences (Wulff et al. 2018). A central challenge in such environments is the exploration-exploitation trade-off. Exploratory behavior aims to acquire information about the environment. For example, prescribing a treatment primarily aims to determine its efficacy for a given individual. In contrast, exploitative behavior seeks to maximize value based on existing knowledge. For instance, it involves prescribing a treatment where the individual efficacy is known (Speekenbrink 2022). Common to exploration setups is that they unfold sequentially within a limited time frame and thus require learning and adaptation. Because exploration and exploitation are often mutually exclusive in the short

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term, exploitation tends to favor familiar options, whereas exploration involves selecting lesser-known alternatives. The extent to which individuals favor exploration over exploitation can thus be viewed as a behavioral tendency, akin to traits such as risk propensity or pro-sociality (Toyokawa et al. 2017). The canonical bandit task is a simple and transparent task that facilitates the study of how people behave when faced with an exploration-exploitation trade-off. In this task, participants try to maximize their total score by repeatedly choosing from several initially unknown options. Picking an option reveals a certain number of points and different options lead on average to more or fewer points. Thus, people need to learn from experience, while their choices are consequential to their score from the outset. When people use an exploratory strategy for this task, they assign a high value to the informativeness that picking an option provides. Therefore, they try out many of the available options and switch between them often. In contrast, an exploitative strategy means that participants decide after a few trials to continuously choose one option.

Despite their ubiquity, exploration–exploitation problems are computationally demanding to solve optimally (Wilson et al. 2021). The success of any strategy in such settings depends on key environmental parameters. Exploratory strategies tend to be advantageous when the decision horizon is long, when the environment is noisy, or when rare but highly rewarding outcomes are present (Sutton and Barto 2018; Teodorescu and Erev 2014). Even in relatively simple and transparent environments, such as canonical multi-armed bandit (MAB) tasks (Thompson 1933), humans often struggle to strike an effective balance, partly due to cognitive resource limitations (Gershman 2020; Laureiro-Martinez et al. 2019; Wu et al. 2018, 2022). In contrast, reinforcement learning (RL) algorithms have demonstrated strong performance in navigating these trade-offs (Sutton and Barto 2018). As a result, exploration–exploitation tasks offer a compelling context in which algorithm advice may augment human performance.

Advice can be any kind of information presented intentionally to human decision makers in order to help them with the decision they are facing (Çelen et al. 2010). While advice can consist of emotional or social support or strategic instructions, research in judgment and decision making often focuses on advice that provides a recommendation of a particular decision option (Bonaccio and Dalal 2006). Following this tradition, we focus on advice that indicates what options should be chosen to enhance the decision maker's performance. Advice can also be differentiated by the source of advice. While many studies on advice focus on advice provided by another human, the literature on advice from algorithms, that is statistical systems that processed data in order to make recommendations, is growing rapidly (Dietvorst et al. 2015; Logg et al. 2019; Chacon et al. 2022; Mahmud et al. 2022; Meehl 1954). This development reflects the growing importance of algorithms as decision support and recommender systems for a variety of decision making tasks. While there is broad consensus that advice, especially from algorithms, has the potential to enhance decision making (Bonaccio and Dalal 2006; Himmelstein and Budescu 2023; Meehl 1954), the effectiveness of such aid depends critically on whether individuals are willing to accept algorithm advice.

Much prior research on algorithmic advice relies on judge–advisor systems (JAS), in which participants must provide a numerical estimate of an objective quantity. For example, participants might be asked to estimate the weight of a person or the popularity of a song (Logg et al. 2019; Sniezek and Buckley 1995; Mahmud et al. 2022; Dietvorst et al. 2015). In these settings, algorithms provide an estimation, which was calculated on the basis of data that was or was not visible for the participants themselves. Despite the clear advantages such simple settings provide (Himmelstein and Budescu 2023), JAS settings cannot account for advice taking and decision-making in DfE settings, since these outcomes of actions are uncertain and the set of observations one can rely on depends on prior exploration behavior. Since algorithm advice becomes more and more broadly used in everyday life, and the situations in which algorithms are used include complex and dynamic decisions such as career decisions (Patil et al. 2023), health related decisions (Malmir et al. 2017) and even moral decisions (Köbis et al. 2021), research on algorithm advice needs to reflect the complexity of the settings in which algorithms are employed. This research addresses the existing gap by aiming to provide insights into how people use algorithmic advice in a bandit task, which represents a dynamic exploration–exploitation scenario. Specifically, we aim to determine which features of the algorithm and the human decision-maker increase the uptake of advice.

Specifically, we focus on how the exploration tendency of algorithms and individuals impacts the willingness to accept algorithmic advice. This is especially relevant since humans differ widely in their exploration tendency (Danwitz and von Helversen 2025) and might benefit greatly from advice that helps them find a good balance. Moreover, successful exploration behavior is not only about finding a balance of exploration and exploitation but also about finding a beneficial exploration strategy (Wilson et al. 2014; Sutton and Barto 2018): While generally an exploration strategy, in which choices are directed to especially informative options is successful, human decision makers often rely on random exploration strategies, in which they randomly select uncertain options (Wu et al. 2022; Brown et al. 2022). At the same time, it is yet very unclear, whether people are open or reluctant to use advice from explorative algorithms and if this depends on their own exploration tendency. Last but not the least, designers of recommendation systems, such as those that propose products in online shopping, face the relevant challenge of designing these systems to balance the exploration–exploitation trade-off (Adamopoulos and Tuzhilin 2014): While users like to get advice that is explorative, in the sense that it was not obvious, they do not like advise that diverges to strongly from their expectations and is thus to some degree exploitative.

In the following, we briefly introduce current general findings on algorithm advice to elaborate on why one should assume a preference of human decision makers for either explorative and unexpected or exploitative and consistent advice and in which ways the similarity in the exploration tendency of human and algorithm might come into play. In addition, we will review literature from human-to-human advice taking, given that there is little literature on algorithmic advice in dynamic exploration–exploitation tasks. While taking advice from an algorithm is often not the same as taking advice from humans (Madhavan and Wiegmann 2007; Dietvorst et al. 2015; Logg et al. 2019),

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the principles of when advice is accepted in human interaction may serve as a blueprint to generate hypotheses about human reactions to algorithms. For example, humans do not only represent the reliability of algorithms or machines but trust or distrust them, and this trust is, among other factors, influenced by reputation and familiarity, similar to trust between humans (Madhavan and Wiegmann 2007). Such parallels of human–human interaction and human–algorithm interaction likely occur when people transfer their experience from interactions with humans to situations with algorithms advice (Liu and Wang 2025).

1.1 | Antecedents of Taking Algorithm Advice

Unsurprisingly, people are generally more willing to use the advice of an algorithm if the algorithm performs well (or is perceived to do so) and the expertise of the advised person is low (e.g., Logg et al. 2019; Saragih and Morrison 2022; Van Swol et al. 2018). However, according to what Madhavan and Wiegmann (2007) refer to as the *perfect automation scheme*, humans expect algorithms to work perfectly, and adherence to an algorithm's recommendations decreases rapidly once the recommendations are perceived as imperfect. This may lead to algorithm aversion in settings or tasks in stochastic environments, such as ours, where it is impossible to provide perfect recommendations (Dietvorst et al. 2015; Dietvorst and Bharti 2020; Prahl and Van Swol 2017). Then again, Filiz et al. (2021) have shown that algorithm aversion decreased when participants had the opportunity to evaluate the algorithm based on feedback over repeated decisions. In a study of Chacon et al. (2022), people even show initially very high reliance on algorithm advice. This study points out that the first impressions people form about algorithms are especially crucial. In the current study, participants are informed that the advice comes from state-of-the-art algorithms, and the algorithms are designed to perform well enough to benefit most users. Therefore, we assume that people will use the algorithmic advice to some degree across all conditions. This assumption also serves as a prerequisite for our study, as we aim to determine the extent to which people rely on algorithms that use different exploration strategies.

Algorithms can be designed to favor exploration or exploitation (Sutton and Barto 2018). Thus, another potentially influential characteristic of the algorithm is its tendency to explore or exploit, as well as how such a preference evolves, if there is one. Directed exploration describes the tendency of humans or algorithms to select options that are especially informative, meaning options that have the potential to significantly reduce uncertainty (Wilson et al. 2014; Chakroun et al. 2020). Algorithms relying on directed exploration could be attractive as people oftentimes fail to employ directed exploration and instead rely on random choices when exploring (Wu et al. 2022). Reliance on such a random exploration strategy in humans is related to limited cognitive resources (Dubois and Hauser 2022; Brown et al. 2022), and overreliance on random exploration has even been associated with mental illnesses such as ADHD and schizophrenia (Addicott et al. 2021; Cathomas et al. 2021). Further, algorithms relying on directed exploration are likely to make suggestions that are less expected by the human decision-maker than exploitative algorithms. Thus, in observational learning, people have the ability to copy explorative agents in a way that helps to reduce their costs for individual exploration (Witt et al. 2024; Danwitz and von Helversen 2025). Also, from the perspective of an individual human decision-maker, explorative advice might appear more informative: While exploitative advice only indicates to exploit, explorative advice also indicates, where precisely to explore.

In contrast, algorithms with a stronger tendency to exploit are by design more consistent over time in their choices. While directed exploration is also a consistent strategy, advice takers might not be capable or willing to evaluate the advisers strategy on the strategy level and therefore perceive explorative advice as inconsistent. Ihssen et al. (2016) found that participants were more willing to follow choices that were consistent over time than choices that varied between options. In settings with sequential choices, consistent strategies, as opposed to changing behavior or choices, could be perceived as a signal of expertise (e.g., Falk and Zimmermann 2017; Fehrler and Hughes 2018; Soll et al. 2022), which would make exploitative algorithms appear as having a higher ability. Our goal is to determine whether people generally prefer the explorative or exploitative algorithm. Since previous literature does not provide conclusive evidence regarding the direction of the effect, and there are compelling arguments for both directions, we cannot formulate a directed hypothesis. To gain further insight into whether participants' assumptions about the properties and usefulness of the algorithm relate to their actual choice behavior, we added self-report questions at the end of the experiment.

Exploration scenarios are settings in which choices and experiences are made sequentially. This does not only allow us to investigate situations in which preferences for either of the algorithms evolve. Moreover, this feature of exploration scenarios captures a feature of many real-world applications of algorithm advice, in which people also need to dynamically form an opinion on an algorithm, based on the advice it gave and potentially the outcomes this advice led to. Past research has stressed that the history of interaction between algorithms and humans accounts for the human evaluation and their willingness to use the algorithm in the future: Algorithm aversion evolves after people see how algorithms err (Dietvorst et al. 2015), while histories of positive interactions predict further positive interactions (Filiz et al. 2021; Chacon et al. 2022). If there is a general preference of people for more explorative or exploitative algorithms, it would thus also be important to pinpoint situations in which this effect evolves. Specifically, a positive interaction might occur, when people make initially positive experiences and then start to generally trust the algorithm (Chacon et al. 2022). On the other hand, a negative interaction might evolve if people have a negative initial experience or decide not to follow the algorithm, but the algorithm sticks to its predictions. If people ignore the algorithm, they may lose trust in it and develop a negative perception of it. This dynamic can be examined by determining how often different algorithms recommend options again after the advisee has previously ignored the advice and by assessing the likelihood that such ignored advice will be disregarded again (Charness and Dave 2017). These mutual dynamics might occur when explorative algorithms advise repeatedly to pick novel options, as well as when exploitative algorithms advise repeatedly

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to exploit known options. Hawkins et al. (2023). find that people prefer to copy from others, if they are currently exploiting. However, in their setup, current exploitation indicates quite clearly, that the other player found a beneficial spot in their exploration game. On the other hand, there might be a preference to learn from currently explorative agents, when this helps to reduce the external and internal costs for individual exploration (Witt et al. 2024). We conducted a purely exploratory analysis to determine whether patterns indicating a dynamically evolving mutual ignorance between the advising algorithm and the advisee exist. We also investigated whether these patterns depend on the algorithm's exploration tendency and how a preference for exploration or exploitation relates to a generalized preference for the advising algorithm (Madhavan and Wiegmann 2007)[1]

Besides the properties of the algorithm, there might also be systematic interindividual differences between humans regarding their willingness to follow algorithm advice (Mahmud et al. 2022). Previous research suggests that the inclination to rely on information from others depends on the participants uncertainty (Toyokawa et al. 2017; Kameda et al. 2022). Thus, we assume that participants, which are not successful on their own, are more prone to use advice.

Another potential individual factor that influences the advice–taker's decision to follow the algorithm's advice is the advice–taker's individual tendency to explore. In general, the individual strategy in exploration–exploitation setups seems to impact the decision to follow advice (Pereg et al. 2024; Najar et al. 2020), as well as individual exploration and learning from others. In some cases, both are used to the same end: to gather knowledge on the environment (Witt et al. 2024). Additionally, in animal studies, relations of individual exploration tendency and the willingness to copy other individuals are observable (Mesoudi et al. 2016). While the only attempt to directly show that highly explorative people follow more advice from others failed to show a connection (Toyokawa et al. 2017), they used a two-armed bandit setup in which participants had only a limited need to explore individually or by relying on social cues. Since our setup will present participants with a strong need to explore, we assume that individually explorative decision-makers should generally be more open to using algorithmic advice.

Lastly, rather than the exploration tendency of the algorithm or the strategy of the human, the fit of both may matter for the human's willingness to follow advice (e.g., Adomavicius et al. 2008). Research on advice-taking from humans indicates that people tend to follow advisors they perceive as similar, whether the similarity is superficial, such as sharing a birthday, or more substantial, such as demographic characteristics (Stolper and Walter 2019). Besides similarity in characteristics between the advisor and advice–taker, people are also more likely to take advice if they have a history of making similar decisions (Kramer 1999). Conversely, people are more likely to disregard advice when there is a pronounced difference between the advisor and the advice-taker (Ecken and Pibernik 2016; Yaniv 2004). This effect is found in factual and preferential matters alike (Ihssen et al. 2016; Analytis et al. 2018; Rader et al. 2017). While these effects of similarity and distance are well researched in JAS with human advisors, it is not yet clear whether they generalize to algorithm advice or advice taking in DfE scenarios (Pálfi et al. 2022). The current study explores how similarity in exploration tendency between advisors and advice–takers impacts advice usage.

1.2 | The Current Research

In the online experiment, participants repeatedly performed a stationary MAB task, and the expected reward of an option remained constant throughout the entire session; see Figure 1A. Compared to restless MABs, where the expected reward changes over time, stationary MABs are easier to solve and show much clearer distinction between exploration and exploitation choices (Daw et al. 2006; Gershman 2020; Sutton and Barto 2018; Speekenbrink 2022). Participants first conducted the MAB on their own and then with the advice from a Kalman filter algorithm with exploration bonus (KAE). The Kalman filter is a state-of-the-art RL algorithm that estimates the expected reward and the uncertainty in the expected reward for each of the options (Daw et al. 2006). The KAE recommends the option with the best combination of a high expected reward and high uncertainty, because choosing uncertain options leads to a higher gain in knowledge about the environment than choosing well-known options. We implemented three treatment conditions between participants, which differed in regard to the directed exploration tendency of the KAE advice. Specifically, there was an overly exploratory, an overly exploitative, and a balanced algorithm, as depicted in Figure 1B–E. These algorithms were calibrated to (1) perform well (i.e. better than an average human decision maker) and (2) to achieve similar net rewards in the absence of human decision-making—though the balanced algorithm produced slightly higher rewards, when all three algorithms performed the task alone (without interference from a participant). A key feature of the experimental setup is that the algorithms generate recommendations based on information derived from participants' actual choices. Thus, our experimental design captures a realistic scenario of algorithm-guided exploration in which humans and algorithms truly interact.

2 | Methods

2.1 | Study Overview

We conducted an online experiment using a 10-armed stationary MAB task. Participants aimed to maximize rewards, earning payments based on their performance either independently or with advice from a Kalman filter algorithm with an exploration bonus (KAE) (Chakroun et al. 2020; Daw et al. 2006). We used a similiar Kalman filter model to model the participants' decision processes, enabling a comparison of the exploration tendencies of the participants and the advising algorithm.

The experiment consisted of two phases. In the first, participants performed the MAB without advice to estimate their individual exploration tendencies. In the second, we implemented four between-participant conditions: three with algorithmic advice and one control without advice. The advice treatments featured KAE recommendations with varying exploration tendencies:

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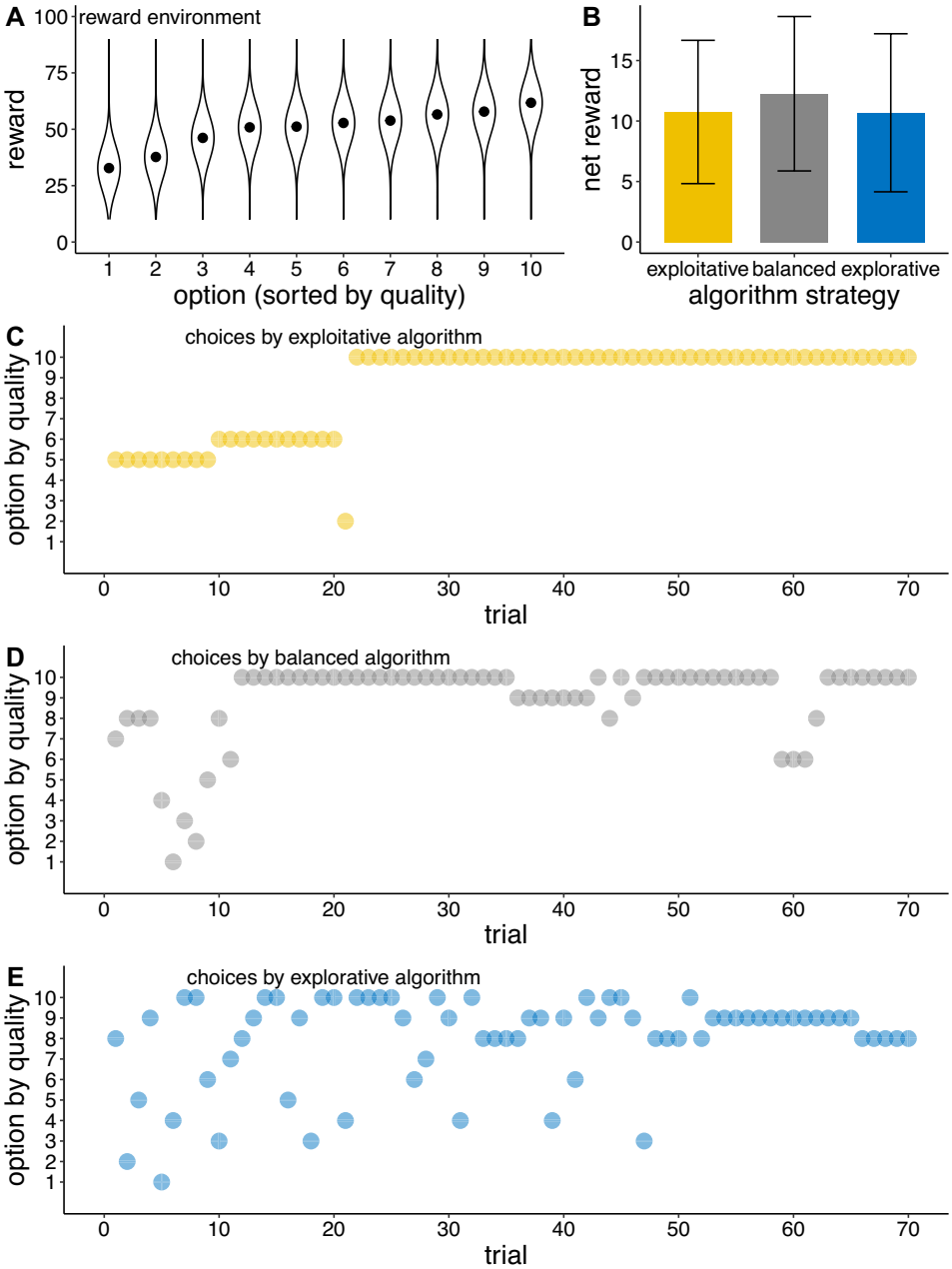


FIGURE 1 | Example multi-armed bandit task and algorithm choices. *Note:* Panel A shows an example of rewards obtained in a multi-armed bandit task (MAB) used in the experiment. For each task, the mean rewards for the option b were randomly drawn from a normal distribution $r_b = \mathcal{N}(\mu = 50, \sigma = 10)$, and rewards were generated probabilistically according to $\mathcal{N}(\mu = r_b, \sigma = 10)$. The violin plots in Panel A give an example of such reward distributions. Panel B shows the means and the standard deviations of the net rewards, that is, the average rewards minus the grand average of 50 points, obtained by the three algorithms over 2000 different MABs. Panels C–E depict the choice sequences of the exploitative, balanced, and explorative algorithms in the MAB illustrated in Panel A, when all three algorithms perform the task alone (without interference from a participant).

explorative, exploitative or balanced. The algorithm's recommendations were based on participants' actual choices and exploration parameters, which fostered genuine human-algorithm interaction.

2.2 | Participants

Participants were recruited through Prolific Academic. Participation was restricted to UK residents to ensure good knowledge of English and comparable value of the study incentives. Participants had to correctly answer comprehension questions about the task goal, reward scheme, and reward generation procedure in order to participate in the experiment. Participants who failed the comprehension questions were removed from the experiment and replaced by new participants until the preregistered number of participants was reached. Six hundred participants finished the study. Among these, 13 have been excluded from the data analyses: 12 because their data were not uploaded due to technical issues and 1 participant because they indicated that they had failed to read the instruction

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regarding the algorithm. Participants' payments consisted of a base payment of £2.50 and a performance-dependent bonus payment. The average bonus was £4.14 per participant. Of the participants willing to provide their demographic details, 65.1% are female, 34.2% are male, and 0.7% are non-binary. The average age is 40.8, with a standard deviation of 23.4 years. Adherence to ethical standards was checked and approved by the German Association for Experimental Economic Research (Certificate No. xBTi83k3).

3 | Bandit Task and Experimental Procedure

Participants performed a total of eight stationary MAB tasks: one training task and three experimental tasks in each of two phases. In the first phase, all participants completed the task individually. In the second phase, all participants except for the control group received algorithm advice. In each task, participants faced 10 boxes and selected one by clicking on it. Upon selection, the box revealed a reward, and participants received feedback on the points earned. The goal was to maximize cumulative points. Payment for participants was based on their average score across the experiment. Box rewards were drawn from normal distributions with means randomly sampled around 50 points (SD = 10) and a trial-level variability of ± 10 points; see Figure 1A. Rewards were rounded to the nearest integer. Each training task consisted of 20 trials, while experimental tasks included 70 trials. Box reward averages remained constant within tasks but were redrawn between tasks, allowing participants to learn which boxes yielded higher payoffs. Participants were informed of trial progress during tasks and received feedback on their average points after each task. The experiment began with informed consent and instructions on the task and procedure. Participants received complete and accurate information about the stochastic process underlying the rewards they would receive (see Supporting Information S4) and about each phase individually. Those in the control group proceeded to the second phase, while those in the algorithmic treatment groups were introduced to the algorithm before the tasks. Depending on the treatment, the algorithm's recommendations were explorative, exploitative, or balanced. The advice was displayed as a yellow square around the suggested box. Participants were rightly informed that the algorithm has received the same information as they have, that is, the information on the stochastic reward environment and the number of points from the boxes picked by the participants but no additional information. After the second phase, the participants were informed of their average points and bonus payment overall. Then, they answered questions about their subjective experience, including reasons for disregarding responses, the effort they exerted, and their reliance on exploration versus exploitation. Those in the advice treatment group also rated the usefulness of the recommendations and their adherence to the algorithm's suggestions using sliders (0 = *not at all*, 1 = *very much*). All participants could leave comments before concluding the experiment. The experiment was programmed in PsychoPy (Peirce et al. 2019) and deployed via Pavlovia.

3.1 | Algorithm Recommendations

To generate the recommendations, we use a Kalman filter algorithm with exploration bonus (KAE) (Chakroun et al. 2020; Daw et al. 2006). In each trial $t \in \{1, \ldots, 70\}$, the algorithm uses the Bayesian learning to generate posterior expectations about the mean and standard error of the mean of the points generated by each box. Let $\hat{\mu}_{b,t}$ denote the posterior expectation of the mean and $\hat{\sigma}_{b,t}$ the posterior expectation of the standard error of the mean of box b in trial t. In the first trial, the expected means and standard errors correspond to the prior expectation, that is, $\hat{\mu}_{b,1} = 50$ and $\hat{\sigma}_{b,1} = 10$ for all boxes. In all subsequent trials, the algorithm updates the expected mean and standard error of the mean of the box selected in the current trial based on the Kalman filter (Daw et al. 2006). The Kalman filter uses the following three recursive equations to update the posterior expectations of the selected box s conditional on the reward R_t obtained by this box in trial t:

$$\begin{aligned} \hat{\mu}_{b,t} &= \hat{\mu}_{b,t-1} + K_t(R_t - \hat{\mu}_{b,t-1}) \\ \hat{\sigma}^2_{b,t} &= (1 - K_t)\hat{\sigma}^2_{b,t-1} \\ K_t &= \frac{\hat{\sigma}^2_{b,t-1}}{\hat{\sigma}^2_{b,t-1} + \sigma^2_{b,0}} \end{aligned} \tag{1}$$

The quantity K_t is called the Kalman learning rate, which regulates how much the expected mean and standard error of the mean are updated based on the expected variance of the selected option. As the expectation for the standard error of the mean equals the known standard deviation of the average rewards before the box is sampled and subsequently decreases, the learning rate starts at one-half and decreases over trials. This implies that the expectation of the standard error of the mean decreases every time the box is sampled, which is a specific feature of stationary bandit tasks. The algorithm recommendation in trial t depends on the posterior expectations $\hat{\mu}_{b,t}$ and $\hat{\sigma}_{b,t}$, and a parameter ϕ that reflects the algorithm's tendency toward exploration. The recommendation for trial t is the box with the largest value of the attraction score $q_{b,t} = \mu_{b,t} + \phi\sigma_{b,t}$. The parameter ϕ controls the directed exploration tendency of the algorithm by scaling the effect of the estimated standard error of the mean on a box's attraction score. If ϕ is large, boxes with uncertain outcomes get a large bonus added on their attraction scores.

We use three different values of the exploration parameter ϕ across the three treatments with algorithm recommendations. In the exploitative treatment $\phi = 0.4$. The balanced treatment uses $\phi = 1.4$ and the explorative treatment $\phi = 3.3$. The three values of ϕ have been selected based on their performance in 10,000 simulated tasks of our experiment. The values used in the exploitative and explorative treatments both generate the same average reward of approximately 61 points across the 10,000 simulated tasks. These values have been selected because they generate an average reward comparable to Bayesian learning in combination with Thompson sampling, which is known to perform well in MAB tasks (Sutton and Barto 2018). The value of $\phi = 1.4$ that we use in the balanced treatment produces an average reward of 62 points across the 10,000 simulation runs, which suggests an optimal balance between exploration and exploitation. Panels B–D of Figure 1 illustrate how the advice by the algorithms in the advice treatments differs by presenting an exemplary sequence of choices when the algorithms perform the task without the human participant.

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3.2 | Computational Modeling

To estimate the exploitation tendency (i.e., the choice consistency) of each participant, we fitted the Kalman Softmax (KSM) model to the first-phase choices. The model uses the same Bayesian learning algorithm we used to generate the recommendations, which returns an attraction score $q_{b,t} = \mu_{b,t}$ for each box. Unlike the attraction score used for algorithm recommendations, the attraction score used to model participants' choices does not include the standard deviation of points generated by the box.[2] Instead of using a deterministic choice rule, we use the *softmax* function with the inverse temperature parameter β as a measure for exploitation. The probability that a participant selects box j in trial t is as follows:

$$Pr(j) = \frac{e^{\beta q_{j,t}}}{\sum_b e^{\beta q_{b,t}}} \quad (2)$$

High values of the individual-specific parameter β indicate an unobserved tendency to exploit the options with the largest expected reward, while values of β near 0 indicate a tendency for random exploration of the environment. Thus, the parameter β captures a participant's unobserved tendency to exploit and can also be interpreted as a measure of the degree of choice consistency. With high values of β, the participant will only switch if the current option turns out to be worse than previously expected but not to randomly explore the environment. We use Bayesian multilevel modeling to estimate the inverse temperature parameter β_1 for each participant in the first phase of the experiment. To assure that the inverse temperature is positive, we use $\beta_1 = e^x$ and a normally distributed prior for the underlying variable x. We report the natural logarithm $log(\beta_1)$ of the inverse temperature in the first phase of the experiment to obtain an approximately normally distributed variation measure of participants' consistency.

To make comparisons between the exploitation tendency of a participant and the exploitation tendency of the algorithm possible, we also fitted the model to the algorithm recommendations. We did this on the individual level for each participant separately to obtain the algorithm's exploitation tendency β_{alg} from the perspective of each participant. We use $log(\beta_1) - log(\beta_{alg})$ as a measure of difference between the exploitation tendency of a participant and the algorithm, with positive values indicating that the participant has a stronger tendency toward exploitation.

For the treatment conditions with recommendations, we use a different model for the second-phase choices. In this model, we do not estimate the random exploration parameter for each individual. Instead, we fixed β to the mean of the posterior parameter distribution of each participant that we estimated based on their first-phase choices. This allows disentangling of the effect of algorithm recommendations from participants' exploration tendencies. To this end, we added a dummy $I_{b,t}$ to the equation of the attraction scores indicating whether the box was recommended by the algorithm, which yields the following:

$$q_{b,t} = \mu_{b,t} + \rho I_{b,t-1} \quad (3)$$

The parameter ρ of this dummy reflects the participant's inclination to follow the recommendations of the algorithm. Positive values of ρ increase the probability that the recommended box is chosen. As in the model of first-phase choices, attraction scores are translated into choice probabilities using a *softmax* with inverse temperature fixed to the estimate of β_1.

The models were fit to the choice data using the Bayesian model fitting software Stan (Carpenter et al. 2017), accessed via rstan (Stan Development Team 2022). For each treatment condition (exploration, balanced, exploitation, and control), we fitted two Bayesian multilevel models: one for the first-phase choices and another for the second-phase choices. Parameter recovery analyses showed that individual model parameters are recovered well from simulated data (see Figure S4A,B). For details on the modeling process and the models properties, see Supporting Information S3 and Table S8.

4 | Results

We analyze the data from the online experiment in two ways. First, we use behavioral proxies for the variables of interest, that is, participants' tendency to explore and their tendency to follow algorithmic advice. We use the number of times participants switch from one option to another as a proxy for exploration, with more switches indicating a higher tendency to explore. Similarly, we use the number of matches with the algorithm's advice as proxy of the participants' inclination to follow advice. Second, since switches and matches can be regarded as approximations but not as valid measures of exploration and advice following, we perform cognitive computational modeling to extract latent parameters for exploration and advice following. We use the net reward of participants as a criterion for performance, that is, the total number of points earned minus the expected points from purely random choices. Additionally, we analyze responses to our postexperimental questionnaire to better understand the motives behind the observed behavior.

4.1 | Behavioral Proxies of Exploration and Advice Following

Performance

Figure 2A shows the net rewards (i.e., the average number of points earned per task minus the expected value of 50 points) participants achieved during the two phases. Overall, participants' performance increased from phase 1 ($M = 57.54$. $SD = 4.64$) to phase 2 ($M = 60.03$, $SD = 4.00$; $F(1, 1172) = 97.37$, $p < 0.001$, partial $\eta^2 = 0.08$; see also Table S1).

In the first phase, in which participants have not yet received advice from an algorithm, we find no significant differences in performance between treatments, ($F(3, 583) = 1.02$, $p = 0.382$, partial $\eta^2 = 0.005$). In Phase 2, net rewards differ significantly between the four treatments, ($F(3, 583) = 6.87$, $p < 0.001$, partial $\eta^2 = 0.03$). This result is due to the comparatively low performance of the participants in the control condition, as the rewards obtained in the three recommendation treatments do

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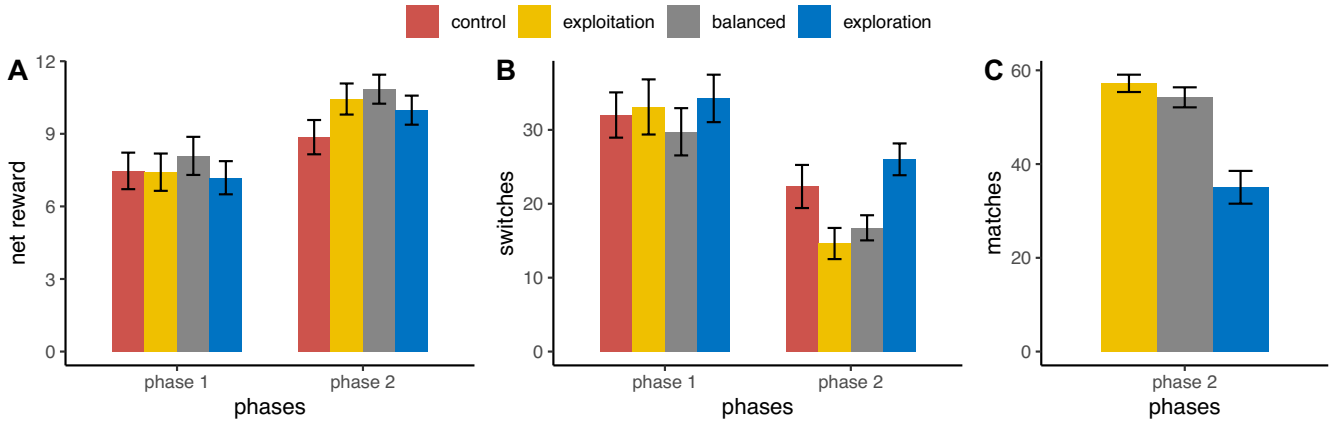


FIGURE 2 | Summary of behavioral proxies by treatment and phase. *Note:* Panel A shows the average net rewards (rewards minus the expected value of 50), Panel B the average number of switches, and Panel C the average percentage of matches between participants' choices and the algorithm's suggestions. All measures are averaged across the three tasks in Phases 1 and 2, respectively. Error bars indicate 95% confidence intervals. Exact values are reported in Table S1.

not differ significantly ($F(2, 439) = 1.93$, $p = 0.147$, partial $\eta^2 = 0.009$).

Paired comparisons between the treatments using the Tukey HSD correction, indicate that, in comparison to the control treatment ($M = 58.9$, $SD = 4.35$), subjects earned significantly higher rewards in the exploitation treatment ($M = 60.4$, $SD = 3.96$; $exploit - control = 1.58$, $p = 0.004$) and the balanced treatment ($M = 60.8$, $SD = 3.72$; $balance - control = 1.98$, $p < 0.001$). Rewards in the exploration treatment were not significantly different from control ($M = 60.0$, $SD = 3.73$; $explore - control = 1.12$, $p = .074$). However, the rewards obtained in the three treatments with the algorithm advice did not differ significantly from each other ($exploit - explore = 0.46$, $p = 0.749$; $balance - explore = 0.87$, $p = 0.233$; $balance - exploit = 0.41$, $p = 0.814$). For all comparisons, see Table S4.

Switches

Figure 2B shows the average number of switches between options in the two phases of the experiment. Participants generally switched more often between options in Phase 1 ($M = 32.28$, $SD = 20.47$) than in Phase 2 ($M = 19.94$, $SD = 14.62$; $F(2, 1172) = 141.3$, $p < .001$, partial $\eta^2 = .11$; see also Table S1). To determine the impact of algorithmic advice on the switching behavior of the participants, we run separate ANOVAs for Phases 1 and 2, comparing the frequency of switches between options across treatments. As expected, there is no difference between the treatment groups in the first phase, in which no algorithm recommendations were presented ($F(3, 583) = 1.31$, $p = 0.271$, partial $\eta^2 = .007$). Regarding Phase 2, we find a significant difference between the treatment groups ($F(3, 583) = 20.47$, $p < 0.001$, partial $\eta^2 = 0.10$) indicating that the recommendations changed how much participants switched between options.

Post-hoc pairwise comparisons between the treatments using Tukey HSD corrections for multiple tests show that participants in the exploitation treatment ($M = 14.6$, $SD = 13.0$) switch less than participants in the exploration treatment ($M = 26.0$, $SD = 13.4$; $exploit - explore = -11.39$, $p < 0.001$) but not compared to participants in the balanced treatment ($M = 16.7$, $SD = 10.6$; $balance - exploit = 2.12$, $p = 0.559$). Participants in the exploration treatment switched more than in the balanced treatment ($balance - explore = -9.27$, $p < 0.001$). In comparison to the control treatment ($M = 22.3$, $SD = 17.9$), participants switched less in both, the exploitation and the balanced treatment

($exploit - control = -7.72$, $p < 0.001$; $balance - control = -5.60$, $p = 0.004$) but switched equally often in the exploration treatment ($explore - control = 3.68$, $p = .110$). For all comparisons, see Table S4.

Matches

Next, we look at how frequently participants' choices matched the algorithm's recommendations, as a rough measure for how often recommendations are followed. As Figure 2C illustrates, the three treatments differ in the average frequency with which participants chose the box that was recommended by the algorithm ($F(2, 439) = 82.26$, $p < 0.001$, partial η^2=0.27, see also Table S1). Tukey tests with HSD corrections show that participants' choices match the algorithm's recommendations less frequently in the exploration treatment than in the other two treatments ($exploit - explore = 22.17$, $p < 0.001$; $balance - explore = 19.19$, $p < 0.001$). Comparing the balanced treatment with the exploitation treatment reveals no significant difference in the frequency of matches between these two treatments ($balance - exploit = -2.98$, $p = 0.252$) For all comparisons, see Table S5.

We conducted follow-up analyses to understand how differences between the algorithms regarding the frequency of matches evolve dynamically. First, we checked how often different algorithms provided advice that was previously ignored by the advisee and how likely such ignored advice was to be ignored again (Charness and Dave 2017). In a second step, we distinguished between cases in which participants had experience with that specific option and those in which they did not.

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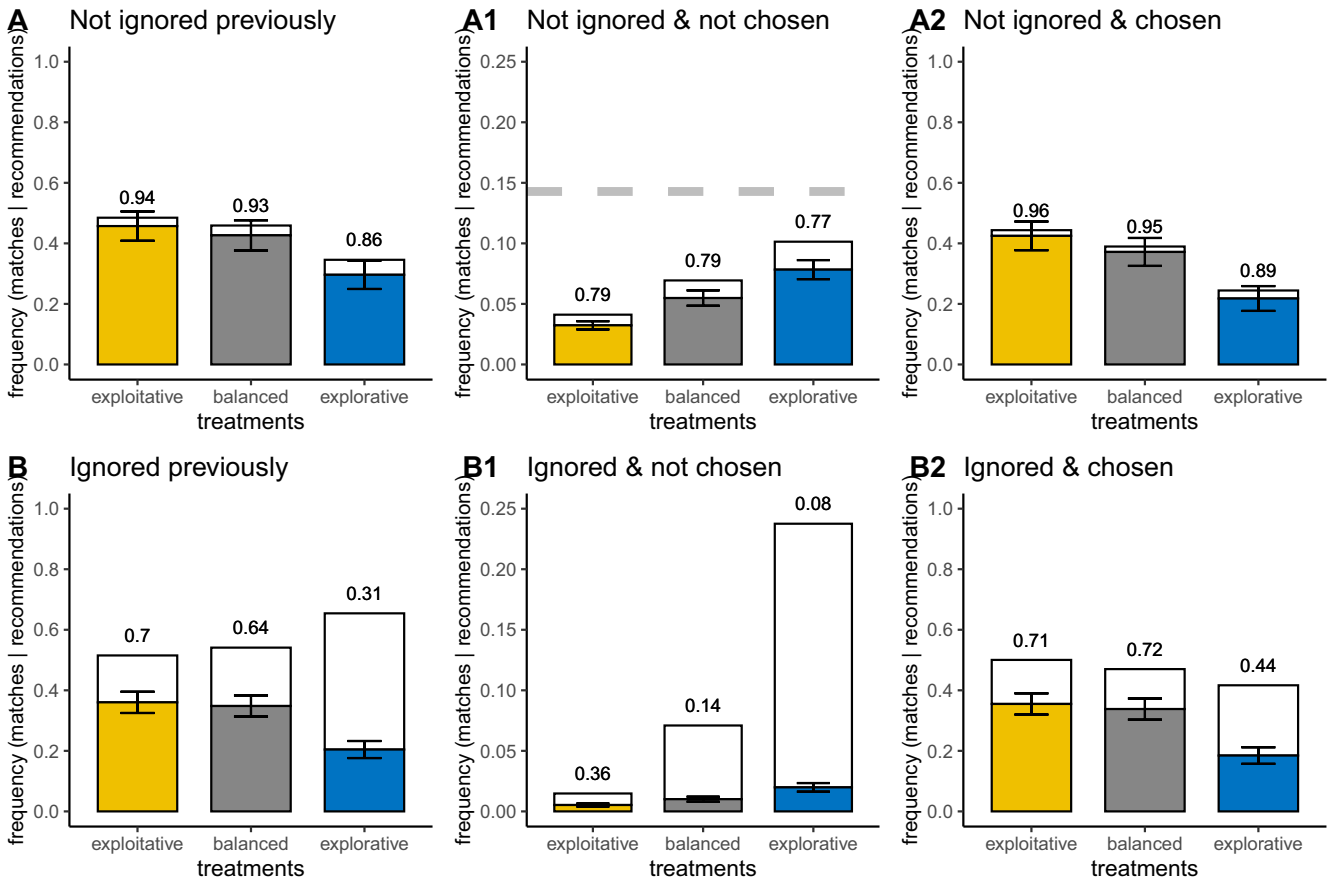


FIGURE 3 | Prior interaction of human and algorithm as predictor of choices matching advice. *Note:* Relative frequency of matches of algorithm advice with participant choice for different conditions. For six types of advice, the total bar (filled bar plus white bar) indicates the relative frequency of some specific type of advice by algorithm treatment. The filled bar indicates how often participants matched with such advice and the white bar indicates how often they did not follow. The number depicts the conditional frequency of matches. For example, in A, the first bar indicates that the exploitative algorithm proposed options which were not ignored yet in ca. 48% of the cases. Of these cases, participants' choices match this advice in 94%. Error bars indicate 95% CI. Panel A1: The gray dashed line indicates that the theoretical maximum for this condition is 10 / 70 cases.

We distinguished cases in which participants had already ignored the algorithm's advice (i.e., a participant did not choose an option that the algorithm advised) or had not yet ignored it (i.e., the algorithm had not yet given this advice or the advice had always been accepted). See Figure 3A,B. As illustrated in the figure, if an option was ignored previously, it was less likely that it was matched afterwards ($t(425) = -15.39, p < 0.001$).[3] The explorative algorithm proposed previously ignored options more often than the two other algorithms ($exploit - explore = t(291.06) = -4.23$, $p < 0.001$; $balance - explore = t(292.51) = -3.40, p = 0.001$). In addition, across situations, in which advice was previously ignored or not ignored, the rate of matches with the explorative algorithm was smaller than the rate of matches with the two other algorithms (advice for previously not ignored options: $exploit - explore = t(289.39) = 5.98, p < 0.001$; $balance - explore = t(287.21) = 5.10, p < 0.001$, advice for previously ignored options: $exploit - explore = t(289.39) = 5.98, p < 0.001$; $balance - explore = t(287.21) = 5.10, p < 0.001$).

We also distinguished between situations in which participants had previously chosen the option proposed by the algorithm and situations in which they had not. We did this to determine the basis on which participants refused or accepted advice. If participants refused advice for options they had never chosen before, we interpreted this as a refusal to try something novel and, thus, a refusal to explore. If participants refused to follow advice for an option they had previously chosen, we can interpret this as a refusal to continue picking or returning to a known option, which is similar to a refusal to exploit.[4]

We further divided these situations into two categories: those in which participants had previously chosen that option and those in which they had not. This resulted in four situations. As expected, more explorative algorithms recommended exploring unchosen options more often. They did this as well, when such advice was beforehand not ignored (Figure 3, Panel A1) ($exploit - explore = t(225.86) = -16.37, p < .001$; $balance - explore = t(291.49) = -7.30, p < .001$), as when the advice to take that option was already ignored (Figure 3B) ($exploit - explore = t(114.49) = -9.98, p < 0.001$; $balance - explore = t(173.15) = -6.57, p < 0.001$). In contrast, more explorative algorithms advised less often to pick a known option when such advice had never been ignored beforehand, approximating situations in which algorithms propose to exploit options (Figure 3A2) ($exploit - explore = t(225.86) = -16.37, p < 0.001$; $balance - explore = t(291.49) = -7.30, p < 0.001$). The explorative algorithm proposed less often to pick options which were

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previously chosen and ignored than the exploitative algorithm; see Figure 3B2) ($exploit - explore = t(266.64) = 2.05, p = 0.041$) but not significantly more than the balanced algorithm ($t(263.81) = 1.06, p = 0.288$). In this situation, the algorithms proposed exploiting a known option for which participants did not consistently follow the advice to choose that option. When the algorithms proposed picking an unchosen option for the first time (i.e., Panel A1), the matching rate was the same for all algorithms. In all other cases (i.e., Panels A2, B1, and B2), the conditional frequency of matching the explorative algorithm was significantly lower than matching the other two algorithms. Thus, participants followed the explorative algorithm less often, even when it proposed exploiting a known option. For details, see Table S6.[5]

Stated Perceptions

In Table 1, we report results from the postexperimental questionnaire to better understand how participants perceived the different algorithms. We asked subjects to rate the usefulness of the algorithm and report their willingness to follow and to rely on the algorithm's recommendations. For more details, see Table S7.

Although we observe no differences between exploitation and balanced treatments, subjects rate the algorithm substantially lower in exploration treatments. They also report a lower tendency to rely on and follow recommendations from the algorithm. See Supporting Information S3 for details[6]

4.2 | Computational Modeling Results

Switches and matches have their limitations as measures of exploration and following: In situations where one box stands out due to large rewards, even a participant with a strong preference for exploration will rarely switch. Similarly, a participant's choice may often match with the recommendation simply because individual experience and recommendations point towards the same option. Furthermore, not every time a participant switches between options indicates exploration. Switches could also indicate returning to a known option after unsuccessful exploration or switching between several options thought to be on the same level. To corroborate the behavioral results, we use cognitive modeling to estimate each participant's latent tendency to exploit, and make consistent choices (β; see formula 2), as well as the latent tendency to follow the recommendations of the algorithm (ρ; see formula 3).

Algorithm Consistency and Advice-Taking

Figure 4A shows that the exploitation tendency of a participant in the first phase is negatively related to the propensity to follow the algorithm's recommendations in the second phase (Pearson's $r = -0.72,\ t(419) = -21.27,\ p < .001$). Participants' who make more consistent choices in Phase 1 (exploit more) follow algorithm advice less. Panel B highlights that the consistency of the algorithm's recommendations also matters. Recommendations that are more consistent are followed more frequently (Pearson's $r = 0.30,\ t(419) = 6.39,\ p < 0.001$), which is in line with participants' self-reported perceptions of the usefulness of the recommendations. The negative slope of the linear fit for the data of the exploitation treatment indicates that the generally positive relationship between advice following and the consistency of advice breaks down at high levels of consistency. As illustrated in Figure 4C, this produces a negative relationship between the estimated latent tendency to follow the recommendations of the algorithm and the difference in consistency between the participant and the algorithm (Pearson's $r = -0.72,\ t(419) = -21.10,\ p < 0.001$). We use $log(\beta_1) - log(\beta_{alg})$ as a measure for the difference in consistency between the participant and the algorithm, with negative values indicating that the participant explored more in Phase 1 than the algorithm recommended in Phase 2. The latent tendency to follow recommendations is stronger for participants who explored more in Phase 1 than the algorithm recommends in Phase 2 ($M = 0.62,\ SD = 0.94$) compared to participants who explored less ($M = -0.61,\ SD = 0.85,\ t(200) = 12.51,\ p < 0.001$).

We use the linear regression models displayed in Table 2 to disentangle the effects of similarity and consistency on the latent tendency to follow algorithm advice. To this end, we regress the natural logarithm of ρ on the negative absolute differences in the exploitation tendency of the participant and the algorithm $-|log(\beta_1) - log(\beta_{alg})|$, serving as a measure of similarity, the consistency of participants' choices in phase one $log(\beta_1)$, and the consistency of the algorithm recommendations $log(\beta_{alg})$. Models (1–3) show that similarity alone does not explain the tendency to follow algorithmic advice well ($R^2 = 0.257$). Moreover, the effect of similarity is substantially smaller and insignificant when controlling for the exploitation tendency of participants, labeled consistency $_{sbj}$, in Model (2). Model (3) shows that including the

TABLE 1 | Summary of postexperiment questions.

Treatment	Algorithm usefulness	Algorithm relying	Algorithm following
Control			
Explorative algorithm	0.50 (0.31)	0.45 (0.31)	0.54 (0.29)
Balanced algorithm	0.67 (0.25)	0.62 (0.28)	0.67 (0.26)
Exploitative algorithm	0.69 (0.21)	0.65 (0.24)	0.66 (0.25)

Note: These are the descriptive statistics of the answers to the postexperiment questions for each treatment (each measured on a scale from 0 to 1). Standard errors are reported in parentheses.

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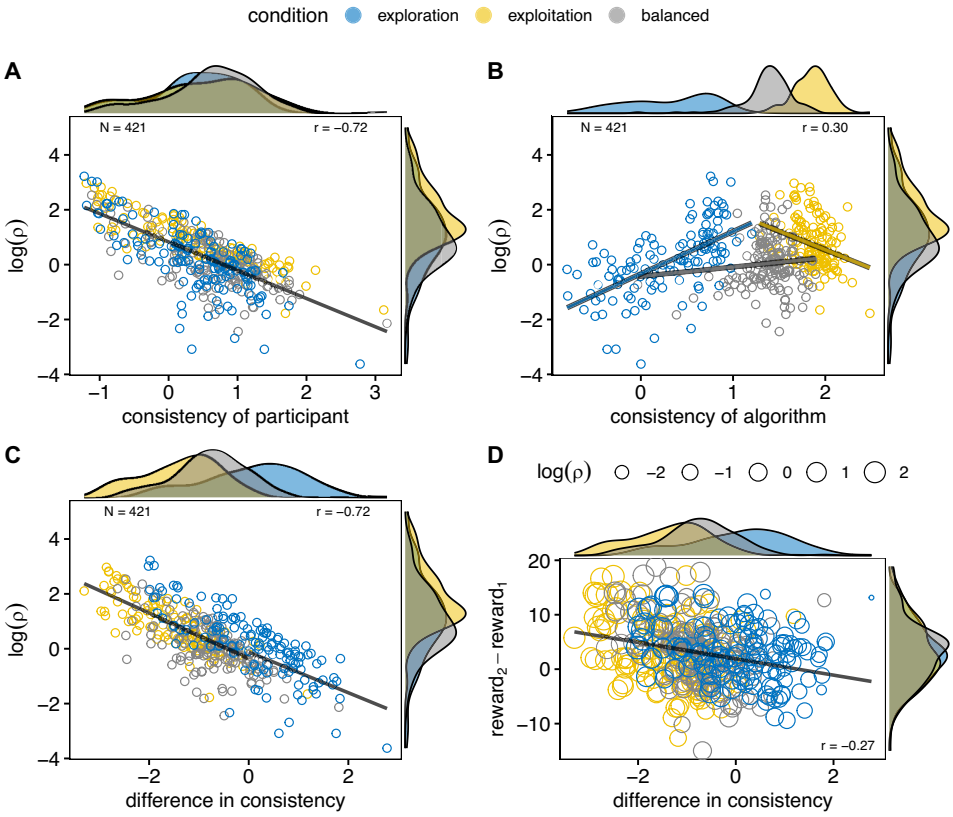


FIGURE 4 | Similarity and Consistency and the Inclination to Follow the Algorithm. *Note:* The figure shows the relation of the natural logarithm of the latent tendency to follow recommendations ρ and the exploitation tendency of the participant and the algorithm, with higher values indicating stronger exploitation (i.e., more consistency). Panel A compares the consistency of participants' choices in the first phase $log(\beta_1)$ with the propensity to follow the recommendations. Panel B relates the tendency to follow the recommendations to the consistency of the recommendations $log(\beta_{alg})$. Panel C relates the tendency to follow the recommendations to the difference between the exploitation tendency of the participant and the algorithm $log(\beta_1) - log(\beta_{alg})$. Separate linear fits are shown for negative and positive differences in consistency to demonstrate that the U-shaped pattern suggested by similarity does not exist. Panel D relates the differences in reward obtained in the second phase compared to the first phase to the difference between the exploitation tendency of the participant and the algorithm. Colors indicate the different treatment conditions with recommendations. The size of the dots is a continuous metric that reflects the tendency to follow algorithmic advice ρ. Parameter distributions are shown on the opposite axes respectively.

consistency of the algorithm $log(\beta_{alg})$ instead of similarity explains the latent tendency to follow algorithmic advice better ($R^2 = 0.611$). Also note that if similarity played an important role, we would expect the regression line shown for the negative differences shown on the left in Figure 4C to have a positive slope, since differences close to zero should produce the strongest tendency to follow algorithmic advice. This is not the case. Both linear regression lines have a negative slope. The corresponding regression analyses relying only on behavioral measures are reported in the Supporting Information S2, Figure S1 A–C, and Table S2.

Individual Consistency and Advice-Taking

Based on the individual parameter estimates, we can also answer the question of who performs best and who benefits most from recommendations. The exploitation tendency of the participant in the first phase $log(\beta_1)$ is substantially correlated with the reward obtained by the same participant in the first phase (Pearson's $r = 0.64,\ t(440) = 17.42,\ p < 0.001$): The more consistent participants are individually (high values of $log(\beta_1)$), the better they perform. Most participants gain more in the second phase of the experiment with the help of recommendations, independent of the treatment condition. Participants who explored more in Phase 1 than the algorithm recommended in Phase 2 benefit the most from receiving recommendations (Pearson's $r = -0.27,\ t(440) = -5.86,\ p < 0.001$). This is shown in Figure 4D. As pointed out above, these participants are also the ones who exhibit the largest latent tendency to follow the recommendations, measured by $log(\rho)$, indicated by the transparency of the points depicted in Figure 4D. For more details, see Figures S2A,B and S3A–C.

5 | Discussion

This study investigated how human behavior in an exploration-exploitation setting evolves when participants receive advice from algorithms with varying exploration tendencies. Overall, we observed that participants incorporate the algorithm's advice, leading to performance improvements and variation in exploration caused by algorithmic advice. In this regard, our

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TABLE 2 | Regressions of $log(\rho)$ on similarity and consistency.

	(1)	(2)	(3)
Similarity	-0.744^{***}	-0.086	
	(0.063)	(0.066)	
Consistency of participant		-1.000^{***}	-1.046^{***}
		(0.064)	(0.045)
Consistency of algorithm			0.420^{***}
			(0.045)
Age	0.002	0.002	0.002
	(0.002)	(0.001)	(0.001)
Male	−0.042	0.165^{**}	0.169^{**}
	(0.095)	(0.076)	(0.070)
Nonbinary	−0.060	0.143	0.276
	(0.654)	(0.520)	(0.474)
Constant	-0.522^{***}	0.589^{***}	0.201^{**}
	(0.113)	(0.114)	(0.085)
Observations	419	419	419
R^2	0.257	0.532	0.611
Adjusted R^2	0.250	0.526	0.606

Note: Linear regressions of the natural logarithm of the latent tendency to follow algorithm recommendations $log(\rho)$ on similarity measured by the negative absolute difference in consistency $-|log(\beta_1) - log(\beta_{alg})|$ between participant and the algorithm, the consistency of the participant's choices in the first phase of the experiment $log(\beta_1)$, and the consistency of the algorithm $log(\beta_{alg})$. Age and gender are used as sociodemographic control variables in all models. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

findings are in line with research showing that when an algorithm performs well (Saragih and Morrison 2022) and people have the opportunity to gain experience with it, they are willing to follow it (Filiz et al. 2021).

In both, the behavioral and computational modeling results, we found strong evidence for a general preference to follow the advice of more exploitative algorithms. Our findings revealed that this preference for exploitative algorithms can be partially explained by the explorative algorithm's tendency to propose so far unexplored options, even if participants already ignored this advice beforehand. This tendency is not a feature of the specific design of the explorative algorithm, but more generally an inevitable feature of explorative advice: If the algorithm would stop to propose unexplored options, it would no longer be explorative. When algorithms advised unexplored (but previously ignored) options, participants followed this advise only very rarely. Conversely, when algorithms advised to take options participants already knew, the rate of matching this advice was higher, even when this advice was previously ignored. In conclusion, participants were more likely to accept exploitative advice. This finding is in line with findings from observational learning: In the setup of Hawkins et al. (2023), current exploitation can be a signal that observed agent found a spot delivering plenty of resources, while exploration might signal that this agent did not find a good spot yet. While this behavior is adaptive in observational learning, in our case, this preference for current exploitation is not necessarily adaptive but might rather count as an instance of a confirmation bias (Charness and Dave 2017; Palminteri and Lebreton 2022). However, this dynamic can only partially explain the preference for exploitative algorithms. Participants also matched recommendations of the explorative algorithm less often than recommendations of the exploitative algorithm, when they were advising to exploit a known option. This evidence for a general preference for the exploitative algorithm over the explorative algorithm is supported by our computational modeling results. These analyses controlled for the effects of the coincidental overlap between individual choice preferences and algorithmic advice by modeling the weight that participants assigned to algorithmic advice. One explanation for this result could be that consistency is regarded as a signal of expertise (Falk and Zimmermann 2017; Fehrler and Hughes 2018; Soll et al. 2022). Following this line of reasoning, participants may have hesitated to follow recommendations to explore because they interpreted exploration as randomness or a lack of expertise demonstrated. In this regard, our findings align with previous research showing that humans rapidly lose trust in algorithms once they observe them making errors (Dietvorst et al. 2015) and participants are especially reluctant to trust algorithms in stochastic environments; hence therein, worse-than-optimal results are inevitable (Dietvorst and Bharti 2020). The importance of such dynamic effects of the history of experience between human and algorithm obtained additional support by exploratory computational modeling analyses; see Figure S2.[7]

In the broader picture, regarding the impact of machine behavior on society, Brinkmann et al. (2023) outlines the optimistic view that algorithms can contribute to the diversification of cultural traits. Modern algorithms and AI not only reproduce what they have learned in the past but also generate entirely new contributions, such as artwork. Technically, they achieve this through a combination of RL and exploration. In our setup, the algorithm that gives recommendations used both of these properties. However, if human advice–takers misunderstand exploration as errors and only accept algorithmic advice that aligns with their preconceptions, the variation stemming from this source may be overlooked. Thus, sequential reliance on algorithmic advice could lead to a biased or narrow information situation, as exploitative advice potentially does in our case (Rahwan et al. 2019).

Focusing on interindividual differences, we found that participants with a strong tendency to random exploration were more likely to follow algorithmic advice. In our sample, a high tendency to explore was associated with poor performance. These findings can be interpreted in two ways, which both are in line with research on social learning strategies of humans and animals (Laland 2004). On the one hand, the finding that especially explorative humans had a higher willingness to follow advice might reflect a general tendency of explorative humans to be open to social advice (Mesoudi et al. 2016). A possible mechanism explaining that tendency considers the individuals' ability to incorporate newly learned aspects of the environment into their mental representation of the environment. If learning from others and learning from individual exploration relies the same

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ability, these tendencies should co-occur in the same individuals (Toyokawa et al. 2017; Pereg et al. 2024). On the other hand, in our sample, explorative behavior is associated with poor performance. It is possible that especially explorative individuals followed the algorithm more because they could benefit more due to their poor initial performance. Therefore, our finding may reflect the tendency of decision-makers to use observational learning when their individually gathered information is inconclusive (Danwitz and von Helversen 2025; Laland 2004; Toyokawa et al. 2017).

We did not find evidence of an impact resulting from the alignment of an individual's exploration preferences and the advising algorithm. This alignment reflects the similarity between the advising algorithm and the advisee regarding their individual preferences for tackling the exploration–exploitation trade-off. While we find a significant effect of similarity, when considering *only* similarity as a predictor, we further show that the inclination to follow advice is better explained by the combination of the algorithms' and the individuals' exploration tendencies. Previous research that has linked advisor and advisee similarity to advice acceptance has mainly focused on situations where people take advice from other people (Yaniv 2004; Yaniv et al. 2011), raising the question of whether such findings generalize to advice acceptance from algorithms (Pálfi et al. 2022; Himmelstein 2022). In this regard, comparing treatments with human and algorithmic recommendations may be a fruitful area of future research. This would allow to clearly distinguish whether our negative finding regarding the effect of similarity on advice acceptance is a specific to algorithmic advice. Findings from comparable settings involving human advice differ from the current study's findings regarding the importance of similarity between advisors and advisees (Wood et al. 2016) and in finding preferences for explorative role models (Danwitz and von Helversen 2025; Charpentier et al. 2020). However, more research is needed to assess whether our negative result on similarity between advisor and advisee generalizes to advice from humans in exploration-exploitation settings, is specific to the type of similarity considered (Yaniv et al. 2011), or is due to the sequential nature of our task.

Author Contributions

All authors equally contributed to the conceptualization of the research. **F.D.:** computational modeling, inference statistics, writing corresponding sections. **L.D.:** writing introduction and discussion, reviewing results and methodology. **F.D.** and **L.D.** share the first authorship. **Y.B.:** inference statistics on the behavioral measures, writing corresponding parts. **H.Y.:** experiment, writing methodology. **B.v.H.**, **L.H.**, **S.F.:** acquisition of financial support, reviewing and editing. **B.v.H.**, **S.F.:** project administration, supervision. **B.v.H.** and **S.F.** share the last authorship.

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Data Availability Statement

The data of the experiment and the analysis code for replicating the modeling are deposited at the Open Science Foundation Repository (https://osf.io/xtgq6). The main hypotheses and analyses were preregistered on the Open Science Foundation (OSF) website, https://osf.io/e3hqa. For a list of the points where we slightly deviated from our plan, see Supporting Information S1.

Endnotes

[1] We would like thank two anonymous reviewers, who suggested to examine the question of the dynamic interaction between the decision maker and the algorithm in more detail.

[2] Including the standard deviation and a parameter ϕ for directed exploration, failed to produce interpretable results. The fitted values of ϕ were mostly negative, indicating that participants actively avoided directed exploration, precluding the use of this parameter as a measure for participants' exploration tendencies in the analysis.

[3] Please note that these follow-up analyses are of an explorative nature and examine the differences between the explorative and the two other algorithms more closely. We thus only look at the differences of this treatment to the two other treatments and do not apply an alpha-error correction.

[4] Please note that it is only an approximation to exploitation to look check if the option was picked previously. There might be cases, in which an option was picked already, but is still picked not to exploit it, but to gather further experience with it and thus reduce uncertainty. For a sharper distinction of exploration and exploitation, turn to the modeling section.

[5] In line with our interpretation of the data that subjects find algorithms more useful that give the consistent advice, we find no evidence for treatment differences in the frequency of matches when focusing only on observations when the algorithm proposes a bandit the first time. In these data, there is no repeated advice of the same bandit in the data anymore, and the p-value of an ANOVA for treatment differences is 0.468. When we instead only remove repeated recommendations that were ignored from the data, the rate of following is higher in the exploitation treatment compared to the exploration treatment ($exploit - explore = 0.19, p < 0.001$) and also higher in the optimal treatment compared to the exploration treatment ($balance - explore = 0.18, p < 0.001$). This supports the conclusion that subjects indeed value consistency in recommendations, ruling out that annoyance from getting advice against one's revealed preferences drives our results.

[6] The exact question on usefulness of the algorithm was "How useful were the suggestions from the AI? Please respond by clicking on the slider below." Participants answered by clicking on a continuous slider (range 0 = not useful to 1 = very useful. The difference between the explorative treatment and both other treatments was highly significant, explorative-balanced algorithm: $t(270.44) = -4.53, p < 0.001$, explorative-exploitative algorithm: $t(253.69) = -5.63, p < 0.001$.

[7] In these analyses, we compared the consistency of the algorithms conducting the task on their own to the apparent consistency of the advice the algorithms actually gave. The findings indicate—in line with the findings on matches—that explorative algorithms, which are repeatedly ignored by exploitative humans, continue to provide explorative advice, and thus superficially appear to be less consistent than they actually are.

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